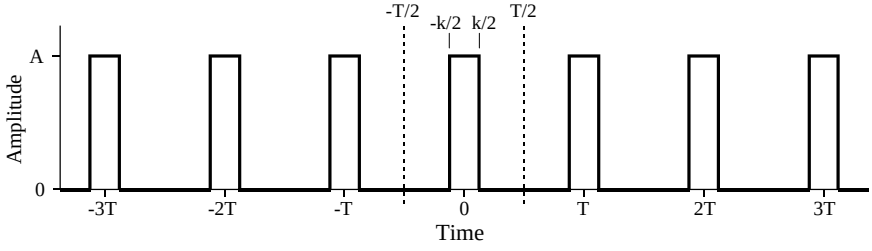

FIGURE 13-11
Example of calculating a Fourier series. This is a pulse train with a duty cycle of $d = k/T$. The Fourier series coefficients are calculated by correlating the waveform with cosine and sine waves over any full period. In this example, the period from $-T/2$ to $T/2$ is used.

Figure 13-11 shows an example of calculating a Fourier series using these equations. The time domain signal being analyzed is a *pulse train*, a square wave with unequal high and low durations. Over a single period from $-T/2$ to $T/2$, the waveform is given by:

$$x(t) = A \qquad \text{for } -k/2 \le t \le k/2$$

$$x(t) = 0 \qquad \text{otherwise}$$

The *duty cycle* of the waveform (the fraction of time that the pulse is "high") is thus given by $d = k/T$. The Fourier series coefficients can be found by evaluating Eq. 13-5. First, we will find the DC component, a_0:

$$a_0 = \frac{1}{T}\int_{-T/2}^{T/2} x(t)\,dt \qquad \text{(start with Eq. 13-5)}$$

$$a_0 = \frac{1}{T}\int_{-k/2}^{k/2} A\,dt \qquad \text{(plug in the signal)}$$

$$a_0 = \frac{A\,k}{T} \qquad \text{(evaluate the integral)}$$

$$a_0 = A\,d \qquad \text{(substitute: } d = k/T\text{)}$$

This result should make intuitive sense; the DC component is simply the average value of the signal. A similar analysis provides the "*a*" coefficients: